

4 (a) $\theta_{rc} \approx \frac{\lambda}{b}$
 $= \frac{1.2}{250}$
 $= 0.0048 \text{ rad} \quad [\text{M1}]$

Using small angle approximations,

$$\theta_{rc} \approx \frac{s}{r} \text{ (arc length formula)}$$

$$0.0048 = \frac{1.8 \times 10^{12}}{r} \quad [\text{M1}]$$

$$r = 3.75 \times 10^{14} \text{ m} \quad [\text{A1}]$$

(b) (i) n can only be odd numbers. [B1]

This is to ensure that the boundary conditions for the formation of a stationary wave within the well is met. [B1]

Or

This is to ensure that a node will be formed at the water surface in the well and an anti-node is formed at the opening of the well. [B1]

(ii) At 30.0 Hz,

$$\frac{n\lambda}{4} = d$$

$$\frac{nv}{4f} = d \quad [\text{M1}]$$

$$(n) \left(\frac{330}{(4)30.0} \right) = d \quad (1)$$

At 42.0 Hz,

$$(n+2) \left(\frac{330}{(4)42.0} \right) = d \quad (2) \quad [\text{M1}]$$

Solving,

$$n = 5$$

$$d = 13.8 \text{ m} \quad [\text{A1}]$$

(iii)



1 mark for solid line and dotted line.

1 mark for shape.

This part can still be awarded if the earlier parts are wrong.